

236502 - Artificial Intelligence project

Presentation

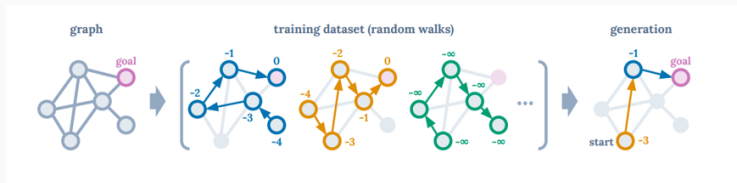
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Decision Transformer for ISO Energy Management

Project Background

- **Background:** In a 2021 research, AI researchers from Barkley, Facebook & Google suggested the DT model for RL problems. In the research, comparisons were made between the DT model to classic RL agents.
- **Decision Transformer (DT)** is a model that uses transformer architecture for reinforcement learning (RL).



- Our goal for this project is to **compare Decision Transformers performance vs. classic RL** in power system control problems.

Decision Transformers (DT) are a recent offline RL approach that frames RL as a sequence modeling task. DT can offer several advantages for power systems control, such as:

- safe offline learning
- long-term dependencies
- flexible objective adaptation
- robustness to sparse or imperfect data

We plan to build and train a Decision Transformer, alongside traditional RL models, and systematically compare their performance.